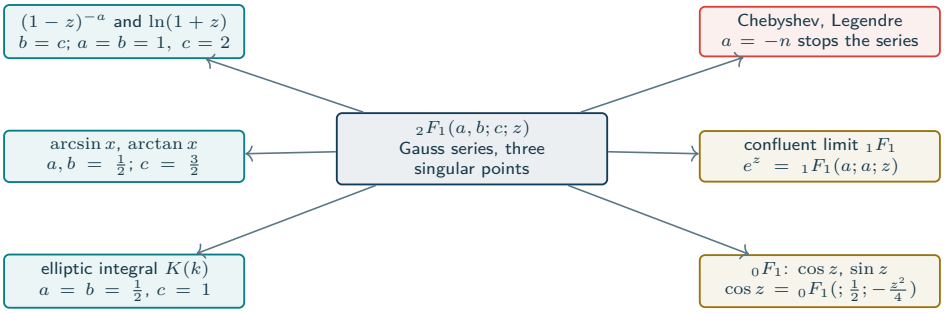
FUNCTIONS LADDER 10 OF 10

Hypergeometric functions

One series with three parameters contains the binomial, logarithm, inverse trigonometric, elliptic and orthogonal polynomial families.

1 / THE RULE AND THE PICTURE

The Gauss series ${}_2F_1(a,b;c;z)$ has many familiar functions as special cases



$$ {}_2F_1(a,b;c;z)=\sum_{n=0}^{\infty}\frac{(a)_n(b)_n}{(c)_n\,n!}\,z^n, \qquad (a)_n=a(a+1)\cdots(a+n-1)=\frac{\Gamma(a+n)}{\Gamma(a)} $$

2 / WHY THIS SERIES / THE RATIO OF TERMS

A series is hypergeometric exactly when the ratio of successive terms is a rational function of n

Here $\dfrac{t_{n+1}}{t_n}=\dfrac{(n+a)(n+b)}{(n+c)(n+1)}\,z$, and for the geometric series the ratio is constant, for e^z it is $z/(n+1)$, and for the binomial series $(n+a)z/(n+1)$, which gives the family ${}_pF_q$ with p factors on top and q below. The series converges for $|z|<1$, and at $z=1$ Gauss's theorem gives the sum when $c>a+b$:

$$ {}_2F_1(a,b;c;1)=\frac{\Gamma(c)\,\Gamma(c-a-b)}{\Gamma(c-a)\,\Gamma(c-b)}, $$

so ${}_2F_1(\frac12,\frac12;\frac32;1)=\dfrac{\Gamma(\frac32)\Gamma(\frac12)}{\Gamma(1)\Gamma(1)}=\dfrac{\sqrt{\pi}}{2}\sqrt{\pi}=\dfrac{\pi}{2}$, which is $\arcsin 1$ from the arcsine row below.

3 / THE IDENTITIES / CHECKED NUMERICALLY

Both sides agree to six figures at the test values

function	as a hypergeometric series	test point	function	series
$(1-z)^{-a}$	${}_2F_1(a,b;b;z)$	$a=2,b=1,z=0.3$	2.04082	2.04082
$\ln(1+z)$	$z\,{}_2F_1(1,1;2;-z)$	$z=0.5$	0.405465	0.405465
$\arcsin x$	$x\,{}_2F_1(\frac12,\frac12;\frac32;x^2)$	$x=0.5$	0.523599	0.523599
$\arctan x$	$x\,{}_2F_1(\frac12,1;\frac32;-x^2)$	$x=0.5$	0.463648	0.463648
$K(k)$	$\frac{\pi}{2}\,{}_2F_1(\frac12,\frac12;1;k^2)$	$k=0.5$	1.685750	1.685750
$T_3(x)=4x^3-3x$	${}_2F_1(-3,3;\frac12;\frac{1-x}{2})$	$x=0.3$	−0.792	−0.792
$P_2(x)=\frac{3x^2-1}{2}$	${}_2F_1(-2,3;1;\frac{1-x}{2})$	$x=0.3$	−0.365	−0.365
$\cos z$	${}_0F_1(;\frac12;-\frac{z^2}{4})$	$z=1$	0.540302	0.540302

For example ${}_2F_1(1,1;2;-z)$ has coefficients $\frac{n!\,n!}{(n+1)!\,n!}=\frac{1}{n+1}$, so $z\,{}_2F_1(1,1;2;-z)=\sum(-1)^n z^{n+1}/(n+1)=\ln(1+z)$, and the first arcsine terms $x+\frac{x^3}{6}+\frac{3x^5}{40}$ come from $(\frac12)_n^2/((\frac32)_n n!)$.

4 / THE DIFFERENTIAL EQUATION / POLYNOMIALS

${}_2F_1$ solves a second-order equation with singular points only at 0, 1 and ∞

The equation is $z(1-z)w''+\big[c-(a+b+1)z\big]w'-abw=0$, and any second-order linear equation with exactly three regular singular points can be brought to this form, which is why so many equations of mathematical physics reduce to it. When $a=-n$ the factor $(a)_k$ vanishes for $k>n$ and the series stops, giving polynomials: Chebyshev $T_n(x)=\cos n\theta$ with $x=\cos\theta$, Legendre $P_n(x)={}_2F_1(-n,n+1;1;\frac{1-x}{2})$ and the Jacobi family.

5 / THE PENDULUM AGAIN / ELLIPTIC INTEGRALS

The exact period is $T=4\sqrt{L/g}\,K(k)$ with $k=\sin\frac{\theta_0}{2}$, and $K=\frac{\pi}{2}\,{}_2F_1(\frac12,\frac12;1;k^2)$

The correction factor is therefore ${}_2F_1(\frac12,\frac12;1;k^2)=1+\frac{k^2}{4}+\frac{9k^4}{64}+\frac{25k^6}{256}+\cdots$, which for a swing of 60° ($k^2=\frac14$) gives 1.0732, the exact ratio of 2.1529 s to 2.0061 s for a 1 m pendulum. For small swings $k\approx\theta_0/2$ and the first term is $1+\theta_0^2/16$, the small-angle correction, so the hypergeometric series organises all the improvements.

ASIDE / THE LADDER IN ONE SERIES

The exponential, the circular and hyperbolic functions, the logarithm and the binomial series are all hypergeometric, differing only in which parameters are set.

LINKS TO OTHER SHEETS

The rising factorial comes from *The gamma function and the beta function*; the elliptic integral appears in *The Second Curve* on the lemniscate; the previous sheet is the gamma function.

Convention. ${}_pF_q$ has p upper and q lower parameters; $(a)_n$ is the rising factorial; K the complete elliptic integral; T_n, P_n Chebyshev and Legendre.

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